

Autocorrelation in Florida weather

November 24, 2025

Data of Florida annual mean temperatures were collected for a century (100 years) between 1901 and 2000. The annual mean temperatures ranges from 23.75 to 26.35 degree Celsius, with a mean of 25.31 (SD = 0.50).

The Pearson correlation coefficient of annual mean temperatures between successive year pairs was 0.32, showing a weak positive autocorrelation. As figure 1 shows, an annual mean temperature is likely to increase as that of its previous year increases. The correlation was statistically significant with an approximate p-value ≤ 0.01 , calculated from permutation analysis (simulation time = 10000).

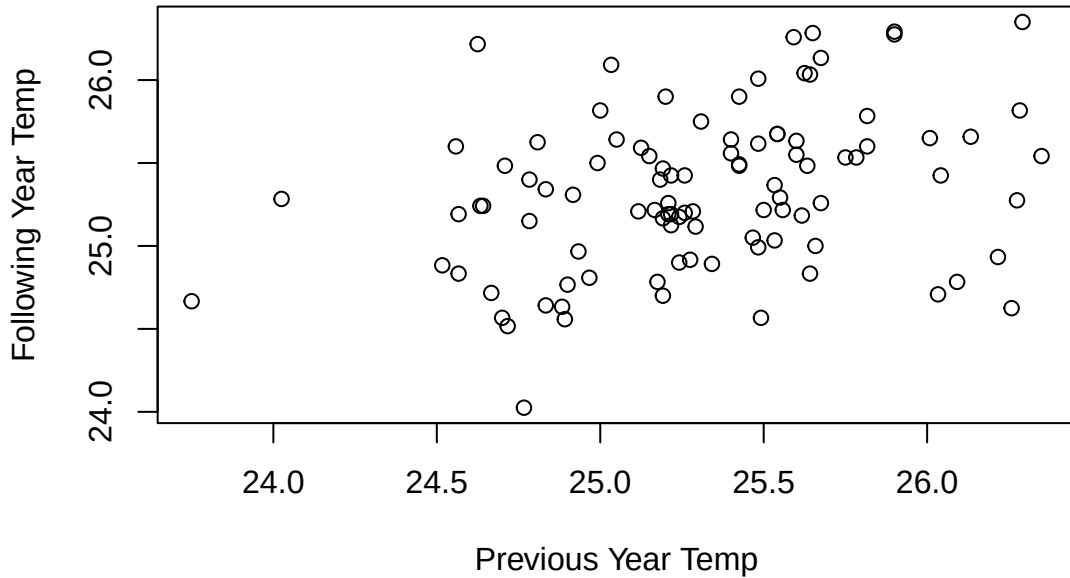


Figure 1: Autocorrelation of successive years in Florida weather. Each point represents the mean temperature (Temp) of a year responding to the mean temperature of its previous year.